

Theme 05: Interruptible Real-Time Agents

Voice-native assistant architecture: fast-and-slow execution and robust interruption handling.

1. Problem Statement & Architecture

Standard AI assistants operate in half-duplex mode (listen, think, speak), failing in full-duplex conversations where users interrupt, re-plan, or correct themselves mid-sentence.

Technically, this is a concurrency and state-consistency challenge. Perception, reasoning, tool execution, and speech must run concurrently on a unified timeline. The standard architecture employs a dual-process model:

- **Fast Path:** Maintains responsiveness within a few hundred milliseconds (acknowledgment, clarification, progress narration).
- **Slow Path:** Executes asynchronous tools, multimodal processing, and complex reasoning.
- **Coordination Layer:** Handles non-blocking execution, call cancellation, state snapshot updates, and idempotency for state-modifying actions.

2. Key Use Cases

- **In-Car & Hands-Free:** Dropping stale route calculations when destination updates mid-execution.
- **Customer Support Bots:** Dynamically adjusting parameters mid-booking without double-booking.
- **Field & Consumer Troubleshooting:** Grounding device queries in camera frames and manuals.
- **Accessibility:** Resolving speech hesitations and conversational self-repairs.

3. Participant Objectives & Interface Contract

Participants implement an agent communicating over two asynchronous queues (timestamped events input, actions output).

3.1 Input & Output Streams

- **Inputs:** Transcribed text chunks (with end-of-turn markers), raw audio clips (WAV), video frames (PNG), interruption signals, asynchronous tool results, and scenario tool manifests.
- **Outputs:** Spoken fillers, non-blocking tool calls (with explicit `call_id`), cancellations, clarification requests, and final responses carrying structured **State Snapshots** (intent and slot values).

3.2 Core Technical Objectives

1. **Floor Management:** Issue meaningful responses quickly without making false completion claims or excessive fillers.
2. **Interruption Recovery:** Promptly cancel superseded in-flight tool calls (within few ms grace period), update state snapshots, and re-plan cleanly.
3. **Session Slot Tracking:** Maintain session-scoped slot state across multi-turn interactions and apply localized slot corrections.
4. **Schema-Driven Tools:** Parse dynamic tool definitions (read-only vs. state-modifying) from manifests; strictly avoid duplicate state-changing calls.
5. **Multimodal Grounding:** Process raw audio and frames behind conversational acknowledgments; clarify ambiguous perceptions.
6. **Protocol Compliance:** Emit well-formed JSON payloads with valid snapshots and identifiers.

4. Evaluation Kit & Mechanics

The evaluation kit mirrors the hidden evaluation environment:

- **Virtual Clock Streaming Harness:** Deterministic event replay, asynchronous mock tool responses, and complete event/action trace logging.
- **Mock Environment:** Deterministic latency and fault injection for flight search, booking, ticket creation, and frame-grounded manual lookups.
- **Public Test Suite:** Nine canonical scenarios (50% text, 30% audio, 20% visual) covering interruptions, chained calls, retries, clarifications, and unseen tools.
- **Hidden Set:** ~60 scenarios (50% text, 30% audio, 20% visual) testing edge cases and adversarial timing.
- **The evaluation kit will be released post the registrations.**

5. Scoring Framework

Each scenario is scored 0–100 based strictly on trace logs:

Category	Weight	Evaluation Criteria
Task Completion	40%	Correct tool execution, valid argument extraction, state snapshot accuracy, and proper final response grounding.
Interruption Recovery	35%	Prompt cancellation of invalidated calls, absence of stale re-runs, and updated state snapshots.
Response Latency	15%	Time to first substantive spoken action following user input or interruption.

Category	Weight	Evaluation Criteria
Safety & Protocol	10%	Zero duplicate state-changing calls, structured schema adherence, and valid state payloads.

A quality multiplier (0.80×–1.20×) evaluates transcript naturalness, truthfulness, and relevance. Hidden scores use a 1.5× multiplier for multimodal scenarios.

6. Execution Constraints & Scope

- **Runtime:** Python 3.10–3.12, 120s wall-clock cap per scenario, 300s setup/warm-up hook.
- **State Scope:** Session-scoped memory only (no cross-session caching).
- **Out of Scope:** Wake-word detection, voice synthesis tuning, UI design.
- **Focus Areas:** Async event loop orchestration, speculative execution, latency hiding, and multimodal intent grounding.

Samsung Electronics